

Implementing modified swarm intelligence algorithm based on Slime moulds for path planning and obstacle avoidance problem in mobile robots

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ABSTRACT

Planning a collision-free path in the least processing time and cost within constraints is a central issue in designing an autonomous mobile robot (AMR). Nature-inspired swarm intelligence (NISI) metaheuristic algorithms are gaining popularity in path planning and obstacle avoidance (PPOA) problem in AMRs. An efficient PPOA algorithm's objective encompasses the ability to read a workspace map and consequently create the shortest collision-free path for the robot to manoeuvre from start to goal in the least processing time and effort. The authors have implemented a modified NISI metaheuristic approach known as a Slime Mould Optimization Algorithm (SMOA) in this research. SMOA takes inspiration from the oscillatory nature of slime mould when it encounters prey. Its mathematical model utilizes adaptive weights to simulate an optimal path for capturing prey or food. The slime moulds produce positive and negative feedback while propagating towards food with excellent exploratory competency and exploitation propensity. For this, simulation has been carried out on MATLAB 2020a. Additionally, the performance of SMOA has been compared with other NISI metaheuristic approaches such as PSO, FA, SFLA and ABC. The results demonstrate that modified SMOA takes less time and effort to generate an optimal collision-free path as compared to other mentioned approaches.

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1. Introduction

In autonomous mobile robots (AMRs), the fundamental problem is to design an optimal and shortest path with the least processing time evading collision with obstacles. Many optimization algorithms have been proposed and implemented by researchers in the last six decades based on the broad application of AMRs. However, due to the dynamic nature of elements in the environment, there is a scope for improvement. Finding an optimal path is a crucial problem in the application of AMRs in industries. With the growing technology trend, automation is the need of the hour. PPOA is an activity that entails a robot to plan or find the shortest collision-free path from one location to another, safely evading collision [1]. The purpose of automation is to reduce manufacturing costs, time and evade accidents. Thus, PPOA is done to generate the shortest collision-free paths within the least processing time and effort [2]. PPOA in robots has been done using several optimization algorithms over the past six decades. However, this is a crucial research domain in robotics

because of the trending demand for automation in manufacturing industries [3].

Problem Statement: To formulate a flexible, robust and easy-to-implement algorithm that can model the virtual environment with precision and can explore the shortest path to achieve a collision-free optimal path in the least processing time and effort.

The PPOA is a non-linear, continuous and non-deterministic problem with several constraints [3]. Due to this character of the PPOA problem, it is difficult to solve it using conventional approaches such as probabilistic road maps, Dijkstra's algorithm, Voronoi Diagram, bugs algorithm and randomly exploring tree, etc. [1,4]. Metaheuristic approaches can be classified into evolutionary algorithms such as simulated annealing and population-based or swarm intelligence based nature-inspired swarm intelligence (NISI) approaches such as particle swarm optimization (PSO), grey wolf optimization (GWO), grasshopper optimization algorithm (GOA) and shuffled frog leaping algorithm (SFLA) [5,6]. NISI algorithms take inspiration from the social behaviour of nature and its elements, such as swarms, flocks, or herds, to generate a globally optimal solution [5]. NISI algorithms' mathematical model is based on a set of rules depending on the swarm and its elements [7]. For example, in particle swarm optimization, the mathematical model is based on particles such

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as fish schooling; in the ABC algorithm, the mathematical model is based on a bee colony [1]. NISI algorithms' objective is to evolve iteratively by substituting present solution with recently generated solution, based on the objective function. The optimization process continues until the maximum number of iterations is reached or the optimal solution is obtained. NISI algorithms are prevalent in the PPOA problem in an existing decade due to better performance, increased efficiency and reduced prerequisite processing time and effort [2]. Conventional approaches lack robustness and randomness, which inclines them to fall into local optimal solutions. NISI algorithms are based on a collaborative approach (parallel-processing) amongst biological clusters and self-organizing capabilities to produce an optimal solution [3]. Therefore, the computational effort and time required to solve an optimization problem using this approach are comparatively less than conventional or evolutionary metaheuristic approaches. All NISI algorithms perform optimization based on two strategies, i.e., exploration and exploitation [7]. The former strategy implies random and exhaustive searching for a possible solution within the entire search space. In contrast, the latter strategy implies précised searching for a possible solution within the neighbourhood. An efficient NISI algorithm aims to solve all optimization problems with less processing time and computational effort and maintain a balance amongst two strategies [8]. Some of the works done in the field of PPOA by the algorithms considered for comparison in this work are PSO [9,10]; FA [7,11–13]; ABC [14–17] and SFLA [18–20].

Methodology adopted for problem solution: The slime moulds belong to *Physarum polycephalum* family and imitate the motion of single-cell organism amoeba [2]. This algorithmic approach has been proposed to elucidate complex multi-objective optimization problems to achieve globally optimal solutions, unlike classical and reactive approaches applicable to the same situation [4]. They have the competence to adapt to changing scenarios to reach nutrients (food or prey), thus being robust and flexible in solving PPOA problems [8]. The purpose of evaluating the PPOA problem using this approach is to achieve collision-free smooth paths. This can be achieved by employing repulsive force from the environment's boundaries and obstacles [4,8]. This work attempts to implement modified SMOA, a NISI-based metaheuristic approach proposed by Li et al. [21] for PPOA problems in AMRs. This work focuses on optimizing and planning an optimal path amongst obstacles.

In previous slime moulds inspired approaches, modelling was done encompassing the complete life cycle of slime moulds. In contrast, the modified SMOA approach, implemented in this work, simulates how slime moulds demean their activities and morphological deviations while executing foraging, thus saving time, effort and energy. Li et al. [21] modified the mathematical model of the SMOA approach by utilizing adaptive weights for simulation. This research article takes inspiration from it to utilize adaptive weights to simulate an optimal path for capturing prey or food so that slime moulds produce positive and negative feedback while propagating towards food with excellent exploratory competency and exploitation propensity. The modified SMOA approach has been successfully simulated on MATLAB 2020a in formulating a collision-free shortest optimal path for a PPOA problem in the AMRs. Simulation is done by outlining five different scenarios (in the form of environmental maps) based on the workspace's complexity. Simulation results demonstrate that modified SMOA performs better than PSO, FA, ABC and SFLA algorithms for similar circumstances in terms of path length and processing time. This research intends to introduce a novel NISI-based metaheuristic algorithm based on simulation results, which has the potential to be implemented in further real-time environments to achieve better decision-making for PPOA problem in AMRs.

This research article is structured in the following manner. Section 2 embodies the contextual motivation and information related to slime moulds and a comprehensive review of the SMOA approach along with its mathematical model. Brief descriptions of the algorithms considered for comparison are discussed in Section 3. Section 4 illustrates the mathematical rules of the SMOA technique to solve the PPOA problem. Section 5 articulates the results obtained after simulation of SMOA in solving the PPOA problem on the MATLAB 2020a platform. Discussion on results obtained is expressed in Section 6. This research article and future scope of improvement with some useful perspectives are stated in Section 7.

2. Slime mould optimization algorithm: Literature review

The SMOA belonged to the *Physarum polycephalum* family and was first proposed by Nagataki et al. in 2000 [2]. The slime moulds have the expertise to generate an optimal path from their inoculation position to the source of the nutrients [8]. Many algorithms have been proposed and implemented based on the *Physarum polycephalum* family, but they suffer from an early convergence issue [2,21]. The modified SMOA algorithm proposed by Li et al. [21] encompasses energy parameters and the basic *Physarum polycephalum* algorithm to obtain optimal solutions in reduced time and number of iterations. *Physarum polycephalum* has characteristics of eukaryotes and can formulate maze problems, theoretical graph-based problems and shortest path problems [21]. The original *Physarum polycephalum* algorithm was tedious to compute for complex problems as it required a considerable number of iterations and a large number of initial parameters [2].

Some of the earlier works encompassing basic versions of the SMOA or *Physarum polycephalum* approach are discussed here. Adamatzky and Jones [22] studied and simulated bio-inspired road planning in urban areas based on foraging tactics followed by *Physarum polycephalum*. They examined and expressed the reconfiguration of transport networks during natural and artificial calamities to minimize damage caused through simulation. When placed in the containment zone, there faced stogy transitions in *Physarum's* foraging behaviour due to its oscillatory tendencies. Liu et al. [3] implemented a *Physarum* optimization approach to solve minimal exposure path planning problem in wireless sensor networks by augmenting existing *Physarum* approach with characteristics such as low complexity and increased parallelism. Zhang et al. [2] proposed an improved *Physarum* algorithm to optimize and generate the shortest path for a path planning problem in wireless sensor networks. They incorporated an energy parameter to reduce processing time and computational effort. Jones and Safonov [4] suggested two strategies to evaluate PPOA problem using SMOA approach. The former strategy employs coupled and non-coupled emergent behaviour which corresponds to foraging and adaptation phenomenon respectively while latter strategy deals with structural technique for evaluating optimization problems using mathematical model. Initially slime moulds dwells into entire workspace which further employs subtractive or shrinkage approach in order to find collision – free path [4,23]. Uemura et al. [24] proposed and simulated an artificial plasmodium algorithm based on an SMOA approach to solve path planning problem in a maze scenario. Wang et al. [25] suggested amoeba based *Physarum* algorithm in which amoeba algorithm was implemented to evaluate shortest path in directed networks and after combining *Physarum* approach, constrained shortest path problem could be evaluated. Guo and Huang [26] implemented *Physarum* approach to model design structure of Integrated Circuits (IC) so as to achieve shorter ICs or simple designs. The literature survey elucidates that the SMOA finds

application in various problem domains such as maze solving problems, graph theory, and path planning problems in wireless sensor networks. All these works lacked in performance compared to the modified SMOA approach proposed by Li et al. [21] in terms of oscillatory tendencies. Modified SMOA algorithm frames slime mould movement towards nutrients while connecting an optimal path into a mathematical model and set of rules to compute PPOA problems in AMRs. When the anterior end of the slime mould extends towards food or prey, the bio-oscillator generates propagating waves. This wave intensifies the stream of slime mould's cytoplasm to encircle and gulp food. The amalgamation of positive-negative feedback helps SMOA to generate globally optimum path and reach comparatively higher goal.

The modified SMOA approach performs by dividing its approach under two heads: foraging and adaptation. In the process of foraging, slime moulds get attracted to nutrients by following diffusion based on chemical gradient. In the adaptation process, they gulp their food or nutrients and modify themselves as per the changing environment [4]. The modified SMOA approach utilizes adaptive weights to simulate an optimal path for capturing prey or food. The slime moulds produce positive and negative feedback while propagating towards food with excellent exploratory competency and exploitation propensity. Implementation of SMOA on PPOA problem has been simulated MATLAB 2020a. The results show that modified SMOA takes less time and effort to generate an optimal collision-free path than other NISI-based metaheuristic approaches such as PSO, FA, ABC and SFLA. The modified SMOA approach frames foraging tactics of *Physarum polycephalum* and not the entire lifecycle in its mathematical model. The modified SMOA approach suggested in this article primarily simulates performance and a morphological deviation of slime mould *Physarum polycephalum* in exploration and does not consider its entire lifecycle. During foraging, organic matter in slime moulds cells search extensively for nutrients in the entire solution space and encircle it by secreting enzymes to engulf and digest it. During the adaptation stage, slime moulds' anterior end prolongs to form a fan shape and move within its solution space to gulp all the nutrients. The modified SMOA approach manifests certain advantages:

(i) The slime mould's weight $\vec{W}$ facilitates in sustaining a specific disturbance rate to ensure a faster convergence rate and circumventing local optima.

(ii) The vibration factor $\vec{V}_b$ permits SMOA to squeeze its cytoplasm in a definite manner, safeguarding efficacy of diversification and accuracy in the intensification phenomenon.

(iii) SMOA improves decision-making capabilities as it can generate the shortest optimal path circumventing obstacles based on information available to it.

3. Brief introduction of algorithms considered for comparison

This section outlines brief description of the algorithms considered for comparison with SMOA approach.

3.1. Artificial Bee Colony optimization (ABC) algorithm

It is a swarm-based NIE approach proposed by Karaboga [27], which mimics the food searching strategy of honeybees. In this algorithm, artificial bees are classified into three classes: employed bees (EB), onlooker bees (OB), and scout bees (SB) [17]. The size of bee colony is twice the number of food positions. Bee colony is composed of EB and OB [14]. After several iterations, if EB cannot improve food position, then it becomes a scout bee. Food searching strategy comprises of (1) EB scan and evaluates nectar quality in search of food; (2) OB receive information from EB and thereby choose better quality nectar as food; (3) SB

then determine locations where food could be found based on motivation or external clues [15,16]. Scouts are regarded as low search costs and a low average in food source quality [14]. Local path searching operation is done by EB and OB, while global path search is carried out by OB and Scout bees [16]. Contreras-Cruz et al. [14] implemented the ABC algorithm for RPP problem in a static environment. ABCO is easy to implement and can be easily modified as per application needs [17]. Its advantages include simple implementation, minimal computation time efforts, minimal processing time, broad applicability, and high flexibility [15–17]. Its limitation comprises of the requirement of new fitness tests on the addition of new parameters. It may lose information, i.e., if the nectar amount of new food source is better than the previous source, then it forgets and memorizes new sources, relatively slow for sequential processes, computation time costs increase with an upsurge in the population [17]. Recent studies have suggested that ABCO is better suited in terms of convergence speed to solve multi-objective optimization problems rather than other evolutionary algorithms like DE and PSO algorithms [28].

3.2. Firefly algorithm

FA presented by Yang (2008) is a bio-inspired (not SI) based nature-inspired algorithm inspired by fireflies' flashing behaviour [7]. Fireflies are winged lighting bugs belonging to the Lampyridae family. Fireflies are characterized by carrying an organic substance, 'Luciferin', which gets oxidized in the presence of enzyme 'Luciferase' to a flashlight [13]. This flashing characteristic (bioluminescence) helps fireflies communicate with one another to attract fellow mates to scare prey [7]. The attraction amongst the opposite gender of fireflies depends on the flashlight's rhythm, rate of flashing light, and time for which flashlight is observed [12]. When FA is implemented for an optimization problem, the objective function is formulated based on fireflies' flashing characteristics [13]. There are two essential issues in the FA approach, i.e., the variation of light intensity and the formulation of attractiveness [13]. Flashing characteristics of fireflies are categorized into the following rules, which form the base of FA [11,13]:

- Fireflies are unisex — i.e., fireflies attract one another despite gender bias.
- Attractiveness and brightness are directly proportional to distance (i.e., brightest fireflies will attract more) and attractiveness and brightness are inversely proportional to distance (i.e., when the distance increases, the impact of brightness and attractiveness reduces).
- The light flashing intensity of fireflies depends on objective function (such that, for maximization, brightness is directly proportional, while minimizing brightness is inversely proportional to the objective function).

This algorithm is mathematically less complicated and avoids erratic flies' movement. This algorithm was integrated with various algorithms (viz. ABCO, GA, neuro-fuzzy, etc.) to enhance further applicability in PPOA problems [13]. Its advantages are: flexible and adaptive, easy to integrate with other algorithms to obtain better results [12], can be employed in complex and hazardous environments [13], minimizes fuel cost, and requires fewer computation time efforts [7]. It performs poorly while varying light intensity and attraction rate for multi-objective problems, limiting its performance [11].

3.3. Shuffled Frog Leaping Algorithm (SFLA)

Food hunting behaviour of frogs inspired Eusuff and Lansey [18] to develop an NIE approach called SFLA. It is a population-based cooperative search method that adopts a frog's natural hunting behaviour to achieve quick and easy applications [18]. SFLA approach is composed of two algorithms in which a genetic-based memetic algorithm performs local search operation and global search is performed by social behaviour based PSO approach [19]. In this approach, a group of virtual frogs is left in a workspace to explore and plan best possible shortest path within minimal processing time [18]. These virtual frogs are partitioned into separate memplexes. Local search operation is performed independently in each memplex [20]. In global search operation, shuffling procedure is done based on a memetic algorithm. This shuffling strategy helps to balance local and global search phenomena giving improved results [19]. SFLA approach can compute complex combinatorial optimization problems in the least possible time with better efficiency [20]. To avoid getting trapped in local minima due to implementation of heuristic approaches for navigation and RPP problem, Ni et al. [20] implemented SFLA to achieve optimal path in dynamic environment. This approach is efficient in handling autonomous navigation and robot path planning compared to other heuristic and NIE approaches [20]. Advantages: Robust approach; can achieve global convergence faster; an optimal solution with higher success rate even in uncertain environment; easy implementation; requires less initial knowledge; avoids local minima [4]; high time complexity with faster search capability [18]; parameter sensitive; more efficient than PSO [19]; minimal computation cost; efficiently handle complex nonlinear combinatorial optimization problems [18]. Limitations: Premature and slow convergence; it falls into local optima when implemented in multi-objective optimization problems [20].

3.4. Particle Swarm Optimization (PSO)

PSO is a swarm-intelligence or population (particles) based nature-inspired optimization algorithm [9] first proposed by Kennedy and Eberhart in 1995. In PSO, robots attempt to mimic the motion of bird-swarms, fish schooling, or animal herds for examining 'collective intelligence' and 'adaptability to environmental changes' for finding food (goal) while escaping predators (obstacles) [9]. Within the population or workspace comprising several particles, a set of particle swarm is randomly initialized, which iterates multiple times based on local and global information gathered from all particles to generate the best optimal path avoiding obstacles or hurdles. In PSO, every particle memorizes its personal best solution (pbest), and the best solution amongst all particles is termed the global best solution (gbest). Compared to other classical algorithms, PSO is simple, easily implementable, robust and non-sensitive to population size. Dewang et al. [10] proposed adaptive PSO to circumvent obstacles while navigating along the path to reach the goal location in a shorter time effectively. In this work, an adaptive PSO algorithm is implemented as it utilizes varying inertia weight (w), which maintains the balance between global and local exploration strategy so that it is high value affects c_2 while low value affects c_1 . PSO has a rapid searching mechanism, and efficiently realizes the problem, optimum for a complex environment. It circumvents local minima problems by increasing population size while maintaining faster convergence speed. Their limitations are [9,10]: falls into local minima for workspace randomly cluttered with obstacles, its performance varies with variation in control parameters and low efficiency.

4. Brief description of mathematical model of SMOA

Many researchers have implemented SMOA or similar approaches to evaluate PPOA problem but the approach suggested in this work contributes a simple approach encompassing behavioural and morphological alterations to mathematically model SMOA approach instead of modelling complete life cycle of *Physarum polycephalum*. In this paper, authors have implemented a modified SMOA approach using different objective function (specified in Section 4.1) for evaluating PPOA problem in AMRs intending to achieve a globally optimal collision-free path in the least processing time with least computational effort and shortest path length. This work is a maiden attempt to implement and perform simulations on MATLAB 2020a using three different morphological alterations of slime moulds to evaluate PPOA problem.

The slime moulds maintain the balance between convergence speed, faster decision-making ability and accuracy to achieve superior foraging capabilities and avoid getting stuck in local minima. The slime moulds adapt to random search situations based on the provenience of food quality [8]. With enriched food's quality, mould will forage within its local neighbourhood and focus on the food source's encircling and with insufficient food's quality; mould will explore alternate search space to find food [2]. The mathematical model of SMOA is based on these criteria. In this section, the mathematical model of the SMOA approach is expressed to elucidate the PPOA problem in AMRs. Refer Li et al. [21] for the detailed concept, background and mathematical model of SMOA. Steps to be followed in the SMOA approach are:- (For definition of symbols expressed in mathematical equations refer Table 1)

A. *Explore and forage nutrients and food sources*: The slime moulds follow food odour in the air to search. Slime moulds are circular in shape. This structure enables them to explore food in all potential directions in the vicinity of the optimal solution. The population of slime moulds is initialized randomly based on Eq. (1). This food searching behaviour is modelled in Eq. (2) mathematically [21].

$$X_i(t) = \text{unifrnd}(LB, UB, 1, N) \quad (1)$$

$$\overrightarrow{X}(t+1) = \begin{cases} \overrightarrow{X_b}(t) + \overrightarrow{vb} + (\overrightarrow{W} \cdot \overrightarrow{X_A}(t) - \overrightarrow{X_B}(t)) \cdot r < p \\ \overrightarrow{vc} \cdot \overrightarrow{X}(t) \cdot r \geq p \end{cases} \quad (2)$$

$$p = \tanh |S(i) - DF| \quad (3)$$

$$a = \text{arctanh} - \left(\frac{t}{\max_t} \right) + 1 \quad (4)$$

B. *Wrap food*: In this step, slime moulds encircle and engulf food source by extending its vein. Eq. (5) computes positive-negative feedback amongst the vein width of slime mould and the quality of food. The searching population (X) at i th iteration is updated with reference to best population location ($\overrightarrow{X_B}$) at i th iteration and its location can be changed by fine-tuning of $\overrightarrow{vb}$, $\overrightarrow{vc}$ and W . The logarithmic factor improves the rate of change of numerical value to circumvent random variation in vein contraction frequency. The slime moulds adjust their exploration process about food quality. When there is sufficient food,

Table 1
Definition of symbols.

S. No	Symbols	Description
1.	$\vec{vb}$	Ranges between $[-a, a]$
2.	$\vec{vc}$	Linear parameter which decreases from 1 to 0
3.	t	Current Iteration
4.	$\vec{X}_b$	Current position with highest odour concentration for food
5.	$\vec{X}$	Location of slime mould
6.	$\vec{X}_A$ and $\vec{X}_B$	Random individuals selected from slime moulds
7.	$\vec{W}$	Weight of slime mould
8.	i	1, 2, ... n
9.	$S(i)$	Denotes $\vec{X}$ fitness
10.	DF	Best fitness obtained in all iterations
11.	rand, r	Random value in the interval of [0, 1]
12.	max_t	Maximum iteration
13.	bF	Optimal fitness at ith iteration
14.	wF	Worst fitness value at ith iteration
15.	SmellIndex	Sort sequence of fitness values (Sort in ascending and descending order for minimization and maximization problem respectively)
16.	LB, UB	Lower bound and Upper bound

then weight obtained within its vicinity holds importance. When food is not sufficient, then weight within its vicinity is reduced, and other areas are explored. The location of the searching individual is given by Eq. (7) [21].

$$\vec{W}(\text{SmellIndex}(i)) = \begin{cases} 1 + r \cdot \log\left(\frac{bF - S(i)}{bF - wF} + 1\right), & \text{condition} \\ 1 - r \cdot \log\left(\frac{bF - S(i)}{bF - wF} + 1\right), & \text{others} \end{cases} \quad (5)$$

$$\text{SmellIndex} = \text{sort}(S) \quad (6)$$

$$\vec{X} = \begin{cases} \text{rand} \cdot (UB - LB) + LB, & \text{rand} < z \\ \vec{X}_b(t) + \vec{vb} \cdot \vec{W} \cdot (\vec{X}_A(t) - \vec{X}_B(t)), & r < p \\ \vec{vc} \cdot \vec{X}(t), & r \geq p \end{cases} \quad (7)$$

C. Capture Prey/Oscillation: The bio-oscillator transmits waves towards better food concentration locations to alter cytoplasmic flow in veins of slime mould. The oscillating frequency in SMOA depends on $\vec{W}$. The oscillating frequency is higher in places with excellent food concentration. Slime moulds reach it rapidly while oscillating frequency is lower in places with meagre food concentration so that the speed of reaching food location is sluggish. This process enhances the proficiency of SMOA and generates a globally optimal solution (high concentration food source). This approach's advantage is that even after obtaining an optimal food source, it would still explore some effort to explore nearby locations for higher optimal solutions to avoid local minima and early convergence errors. The pseudo-code of the SMOA approach is illustrated in Fig. 1.

4.1. Objective function

The solutions of any nature-inspired meta-heuristic algorithm are scrutinized using an objective function. The objective function in this work has been designed to meet two objectives. The first objective is to find the shortest path, while the second objective

is to escape the obstacle collisions. These objectives lead to the formulation of the following objective function:

$$\text{Objective Value (OV)} = PL + PL * \gamma * D \quad (8)$$

where, OV represents objective value, PL denotes path length, γ is the scaling factor and D is the average distance values of all the interpolant points from all the obstacles. It is made zero when the maximum number of interpolant points are lying outside of the obstacles. As indicated in the work objectives, the mentioned objective function is meant to be minimized by all nature-inspired algorithms. This will try to lessen the length of the path, which satisfies the first constraints. The second objective is fulfilled by minimizing D. The scaling factor is there to determine the influence of distance of the path from the obstacles. More the scaling factor, more emphasis will be paid to discourage those paths which contain collisions.

5. Simulation work

The modified SMOA approach proposed by Li et al. [21] was implemented and simulated on MATLAB 2020a for the PPOA problem in AMRs. The performance and efficiency of the SMOA approach were checked and compared with other NISI-based metaheuristic approaches such as PSO, FA, ABC and SFLA algorithm. The simulation work was carried on the 64-bit operating system, windows 10 home single-language laptop and Intel® Core™ i5-4210u processor. The internal memory is 12 GB with 1 TB of the hard disc. For simulation work, four maps were formed with different dimensions and several obstacles. The initial and final coordinates of each map path are specified in Table 2, along with the dimensions and coordinates of obstacles in x and y coordinates. The simulation results obtained are illustrated in this section. While designing an algorithm, specific parameters known as control parameters are assumed at the initialization. The values of control parameters specific to respective algorithms are exemplified in Table 3. For impartial comparison amongst other algorithms, specific common control parameters are initialized with the same values. (See Table 3).

6. Results and discussion

The authors have implemented a modified SMOA approach for PPOA problems in AMRs, intending to achieve a globally optimal collision-free path in the least processing time and shortest path length. Owing to the following characteristics, the modified SMOA approach manifests certain advantages:

- The slime mould's weight $\vec{W}$ facilitates in sustaining a specific disturbance rate to ensure a faster convergence rate and circumvent local optima.
- The vibration factor $\vec{vb}$ permits SMOA to squeeze its cytoplasm in a definite manner, safeguarding efficacy of diversification and accuracy in the intensification phenomenon.
- SMOA improves decision-making capabilities as it can generate the shortest optimal path circumventing obstacles based on information available to it.

This work is a maiden attempt to implement a modified SMOA approach for the PPOA problem in robots. Simulation has been carried out on MATLAB 2020a. In order to implement the algorithm in varied, complex levels, five environment maps were taken for the simulation that were different in obstacles' size, number and locations. After exploring the applicability of modified SMOA in PPOA problems, the results obtained by this algorithm have been compared (for all five maps) with those obtained from other metaheuristic approaches such as PSO, FA, ABC and

SMOA Pseudo code

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Initialize the population size and maximum no. of iteration;
Initialize the positions of slime mould  $X_i$  ( $i = 1, 2, 3, \dots, n$ );
for ( $i = 0, i \leq \text{Max\_Iteration}, i++$ )
    Calculate the fitness of all slime mould;
    Update the bestFitness,  $X_b$ 
    Calculate the W by following equation;

$$W(\text{SmellIndex}(i)) = \begin{cases} \frac{1+r \cdot \log(\frac{bF-S(i)}{bF-WF}+1)}{1-r \cdot \log(\frac{bF-S(i)}{bF-WF}+1)}, & \text{condition} \\ & \text{others} \end{cases}$$

For each search portion
    Update p,vb, vc;
    Update positions by following equation;

$$\vec{X} = \begin{cases} \text{rand} \cdot (UB - LB) + LB, & \text{rand} < z \\ X_b(t) + vb \cdot (W \cdot (X_A(t) - X_B(t))), & r < p \end{cases}$$


$$\vec{vc} \cdot \vec{X}(t), r \geq p$$

End For
End For
Return bestFitness,  $X_b$ ;

```

Fig. 1. Pseudo code of SMOA [21].

Table 2

Type of environment.

S. No	No. of obstacles	Initial coordinates	Final coordinates	X axis	Y axis	Obstacle size
Map 1	3	0, 0	4, 6	[1 1.8 4.5]	[1 5.0 0.9]	[0.8 1.5 1]
Map 2	6	0, 0	10, 10	[1.5 8.5 3.2 6.0 1.2 7.0]	[4.5 6.5 2.5 3.5 1.5 8.0]	[1.5 0.9 0.4 0.6 0.8 0.6]
Map 3	13	3, 3	14, 14	[1.5 4.0 1.2 5.2 9.5 6.5 10.8 5.9 3.4 8.6 11.6 3.3 11.8]	[4.5 3.0 1.5 3.7 10.3 7.3 6.3 9.9 5.6 8.2 8.6 11.5 11.5]	[0.5 0.4 0.4 0.8 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7]
Map 4	30	3, 3	14, 14	[10.1 10.6 11.1 11.6 12.1 11.2 11.7 12.2 12.7 13.2 11.4 11.9 12.4 12.9 13.4 8 8.5 9 9.5 10 9.3 9.8 10.3 10.8 11.3 5.9 6.4 6.9 7.4 7.9]	[8.8 8.8 8.8 8.8 8.8 11.7 11.7 11.7 11.7 11.7 9.3 9.3 9.3 9.3 9.3 5.3 5.3 5.3 5.3 5.3 6.7 6.7 6.7 6.7 6.7 8.4 8.4 8.4 8.4]	[0.4 0.4]
Map 5	45	0,0	15, 15	[4 4 4 4 4 4 4 4 4 6 6 6 6 6 6 6 6 6 8 8 8 8 8 8 8 8 8 10 10 10 10 10 10 10 10 10 12 12 12 12 12 14 14 14 14]	[3 3.5 4 4.5 5 5.5 6 6.5 7 8 8.5 9 9.5 10 10.5 11 11.5 12 1 1.5 2 2.5 3 3.4 4 4.5 5 6 6.5 7 7.5 8 8.5 9 9.5 10 10 10.5 11 11.5 12 10 10.5 11 11.5]	[0.4 0.4]

Table 3

Values of control parameters.

S. No	Parameter name	Parameter values				
		PSO	FA	ABC	SFLA	SMOA
1	Number of handling points (n)	5	5	5	5	5
2	Maximum no. of iterations	500	500	500	500	500
3	Population size (Particles, fireflies, bees, frogs or hawks in respective algorithms)	150	150	150	150	150
4	Inertia weight	1	–	–	–	–
5	Inertia weight damping ratio	0.98	–	–	–	–
6	Personal learning coefficient	1.5	–	–	–	–
7	Global learning coefficient	1.5	–	–	–	–
8	Mutation coefficient rate	0.1	–	–	–	–
9	Activeness coefficient	–	2	–	–	–
10	Light absorption coefficient	–	1	–	–	–
11	Attraction coefficient base value	–	0.2	–	–	–
12	Mutation coefficient	–	0.2	–	–	–
13	Mutation coefficient damping ratio	–	0.98	–	–	–
14	No. of onlooker bees	–	–	150	–	–
15	Abandonment limit parameter or trial limit	–	–	450	–	–
16	Acceleration coefficient	–	–	1	–	–
17	Memeplex size	–	–	–	10	–
18	Number of memeplexes	–	–	–	5	–
19	Number of parents	–	–	–	45	–
20	Number of offsprings	–	–	–	3	–
21	Maximum no of offsprings iterations	–	–	–	5	–
22	Step size	–	–	–	2	–
23	z (Constant)	–	–	–	–	0.3

SFLA, initializing the same values of standard control parameters for unbiased comparison. However, the simulation work has been done with static obstacles, but this approach can also be applied to real-time dynamic systems. The simulation results

show that the modified SMOA approach can generate the shortest path evading obstacles within the path in the least processing time compared to other mentioned approaches. Table 4 clearly explains that the SMOA approach takes less time to compute the

Table 4

Summary of performance of all algorithms in terms of path length and processing time in 5 tests each.

S. No	PSO		FA		ABC		SFLA		SMOA	
	Path length	Processing time	Path length	Processing time	Path length	Processing time	Path length	Processing time	Path length	Processing time
Map 1	9.6431	148.78	9.6315	151.31	8.0678	105.77	8.168	105.63	7.7319	82.25
	11.4621	150.04	10.06	161.52	7.7656	100.93	7.7363	96.98	7.7505	86.69
	9.6387	147.38	10.9853	166.15	8.2958	108.88	7.7327	97.84	7.736	98.03
	11.4076	156.30	10.4937	169.31	9.6356	109.20	7.9398	104.81	7.5184	79.14
	11.0281	151.10	9.6333	158.88	7.7891	104.81	7.7424	103.82	7.737	96.22
Map 2	16.2441	162.43	15.9653	182.41	14.9478	121.10	14.4965	120.94	14.3035	92.53
	16.2424	156.55	16.277	185.18	16.234	132.41	14.9764	126.11	14.3023	91.77
	15.6347	145.66	16.2444	169.20	15.3632	128.07	14.314	116.42	14.5903	96.20
	16.2437	166.46	15.3696	179.44	16.2479	135.89	14.3291	119.07	14.3059	93.18
	15.3649	143.87	15.4885	191.52	15.5531	134.60	15.3566	125.56	14.3074	92.55
Map 3	18.8803	184.28	18.5885	204.79	18.1752	141.94	16.7126	115.34	16.8646	111.17
	19.6852	193.13	18.819	202.42	18.0202	135.45	18.0708	130.57	15.856	98.44
	19.3999	179.54	19.19	216.59	16.8646	130.92	16.811	128.25	17.105	115.67
	18.9573	186.12	19.671	203.45	18.822	148.64	17.0253	131.44	15.8604	102.88
	19.4223	192.65	18.1989	196.58	17.0428	137.47	17.3127	123.69	18.0219	125.83
Map 4	18.1565	208.72	16.2976	223.07	16.2	156.60	16.3033	151.55	16.2927	135.52
	18.4705	215.94	17.8901	233.68	16.2087	160.44	16.1994	148.23	16.1951	132.97
	16.3113	201.08	16.3071	227.15	16.2442	159.53	16.2075	154.20	16.2002	147.23
	17.0362	214.21	18.4118	239.04	16.2943	157.56	16.1963	146.74	16.1994	138.58
	18.0676	210.81	16.3033	235.51	16.3381	168.23	16.2458	149.89	16.2433	134.25
Map 5	21.8651	226.33	21.8553	251.73	21.879	202.73	21.8665	175.28	21.5819	158.73
	21.9283	214.27	21.8964	258.41	21.8848	195.57	21.8679	171.38	21.5556	154.26
	21.9644	206.01	21.9117	263.97	21.7676	198.15	21.7142	164.31	21.863	164.65
	24.6515	219.65	22.2071	261.88	21.9553	206.61	21.7802	179.65	21.8545	162.45
	21.8946	225.46	21.8764	259.09	21.9561	204.39	21.8894	181.46	21.7592	159.17

shortest path, followed by SFLA, ABC, PSO and FA. For each type of environment (five maps), five tests were conducted for 500 iterations to verify the repeatability competence of the SMOA approach. The results represented in Table 4 clearly illustrate that the SMOA approach shows precision and repeatability, unlike other approaches taken for comparison. The best optimal path generated by each approach is shown in Figs. 2 to 6. The percentage change in path length and processing time of SMOA in comparison to other approaches is illustrated in Tables 5 and 6.

Optimal paths generated by all approaches for the first environmental setup is shown in Fig. 2(a–e). PSO takes the longest route to traverse from the initial to the final point, i.e., 9.6387 units in 147.38 s, while SMOA takes the shortest collision-free route, i.e., 7.5184 units in 79.14 s. Compared to PSO, the SMOA approach has the competence to reduce path length by 22%, while the percentage reduction in time taken to compute path is 46.30%. SMOA approach is better than the SFLA approach by 2.77% in terms of path length and 19.11% in terms of time taken for the 1st map. Fig. 2–e illustrates that SMOA circumvents local optima, unlike PSO, FA, ABC and SFLA approaches. The comparison between processing time and best optimal paths generated by each approach simulated for this setup is shown in Fig. 7.

A comparison of optimal paths generated by all approaches for the second environmental setup is shown in Fig. 3(a–e). The simulation results indicate that SMOA outperforms other methods for this setup with the least path length of 14.3023 units in 91.77 s, followed by SFLA, ABC, PSO and FA. SMOA approach is better than the SFLA approach as it takes less time to generate a globally optimal path. For this environment, SMOA takes 6.92% less route and 36.21% less time in traversing from initial to goal point in comparison to PSO. SMOA approach reduces 0.08% of path length compared to the SFLA approach while taking 21.17% less time. The comparison between processing time and best optimal paths generated by each approach simulated for this setup is shown in Fig. 8.

Fig. 4(a–e) shows the best performance of all approaches for the third environmental setup with 13 obstacles of different sizes. Fig. 4(a) displays that during simulation, PSO meets with

Table 5

Comparison of percentage change in path length amongst SMOA and other algorithms.

S. No	PSO–SMOA	FA–SMOA	ABC–SMOA	SFLA–SMOA
	% Change in path length	% Change in path length	% Change in path length	% Change in path length
1	22.00	21.94	3.18	2.77
2	6.92	6.94	4.32	0.08
3	16.02	12.87	5.98	5.13
4	0.71	0.63	0.03	0.01
5	1.42	1.37	0.97	0.73

Table 6

Comparison of percentage change in processing time amongst SMOA and other algorithms.

S. No	PSO–SMOA	FA–SMOA	ABC–SMOA	SFLA–SMOA
	% Change in processing time	% Change in processing time	% Change in processing time	% Change in processing time
1	46.30	47.70	21.59	19.11
2	36.21	48.86	24.22	21.17
3	46.58	49.92	24.81	14.65
4	33.87	40.39	15.09	9.38
5	31.84	38.72	22.15	6.12

a collision during PPOA. Fig. 4(b–d) depicts that FA, ABC and SFLA approach can generate a collision-free path but get stuck in local optima. In contrast, SMOA generates a globally optimal collision-free path of the length of 15.856 units in 98.44 s. The optimal path generated by the SMOA approach is 16.02% less than the PSO approach and 5.13% less than the SFLA approach, which improves its efficiency by reducing incurring costs in designing the shortest path. The SFLA approach takes 46.58% and 14.65% less time than PSO and SFLA, respectively. The comparison between processing time and best optimal paths generated by each approach simulated for this setup is shown in Fig. 9.

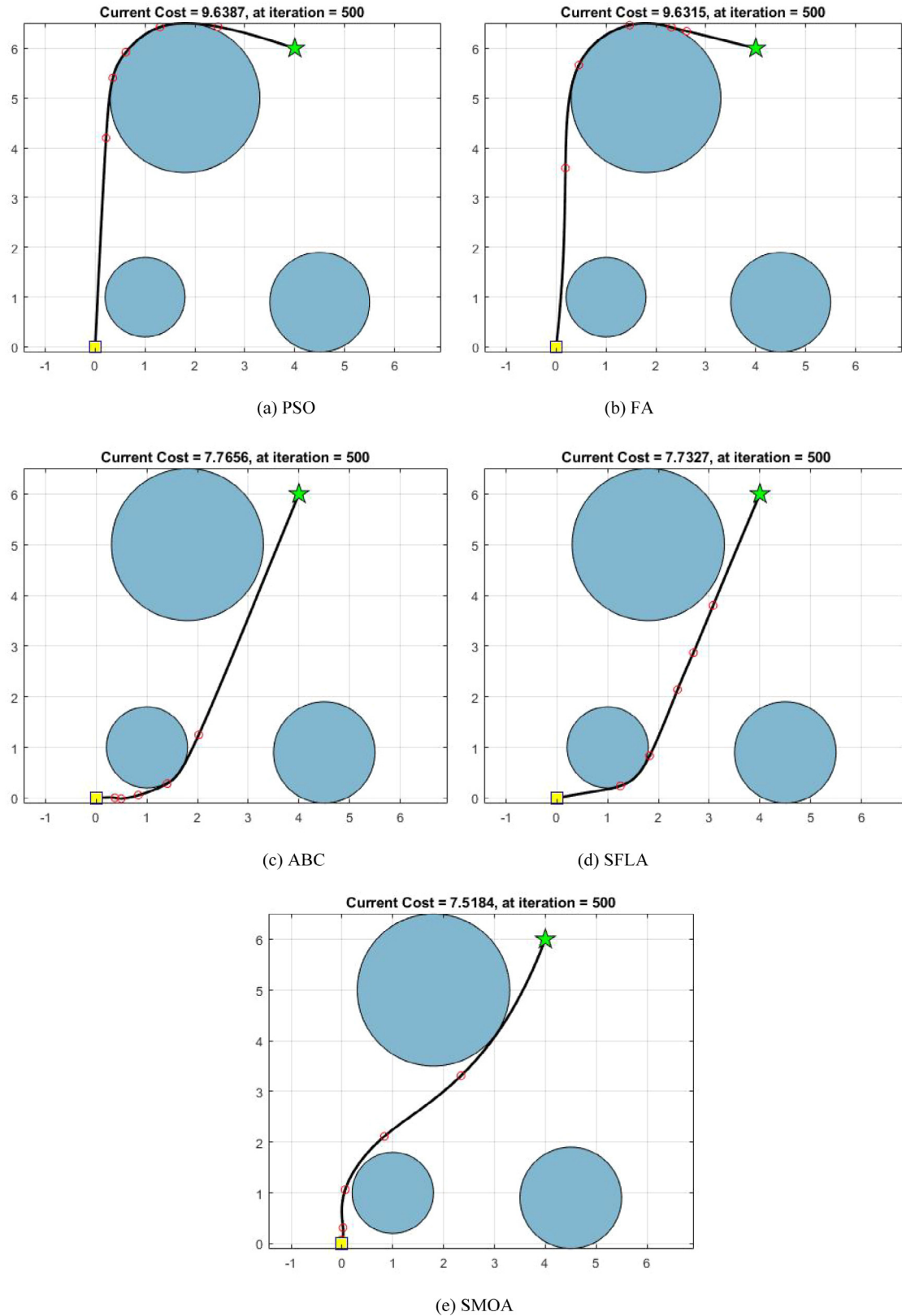


Fig. 2. Map 1 (a) PSO, (b) FA, (c) ABC, (d) SFLA and (e) SMOA.

The fourth environmental setup's simulation results depict that all approaches take the same route or trajectory while manoeuvring from initial to a final position to generate a collision-free optimal path (illustrated in Fig. 5(a–e)). The SMOA approach

takes 132.97 s to forge a globally optimal path of the length of 16.1951 s. The route taken by SMOA is straightforward and smooth compared to the curved path generated by PSO and FA approaches. The SMOA approach requires 33.87% and 9.38%

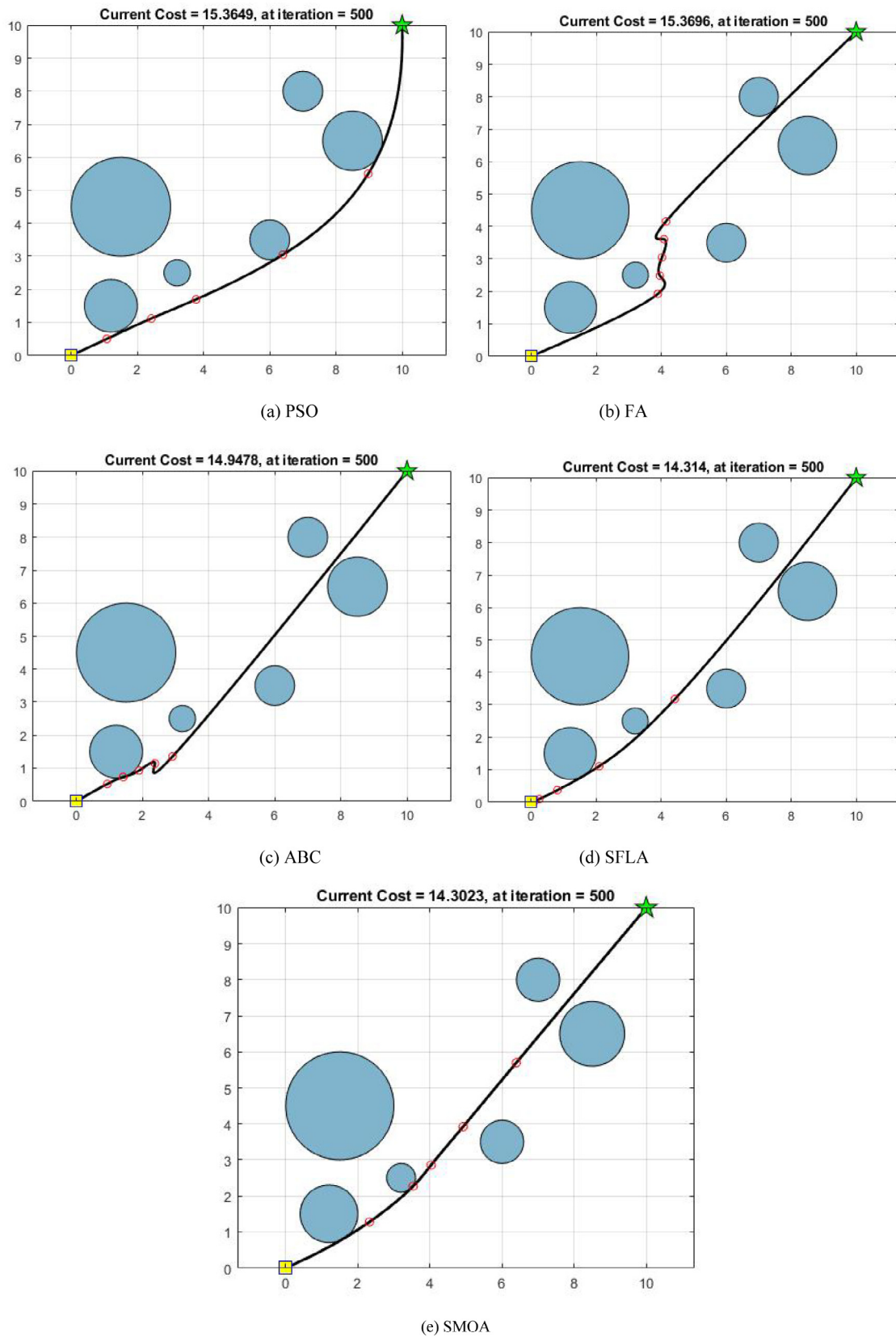


Fig. 3. Map 2 (a) PSO, (b) FA, (c) ABC, (d) SFLA and (e) SMOA.

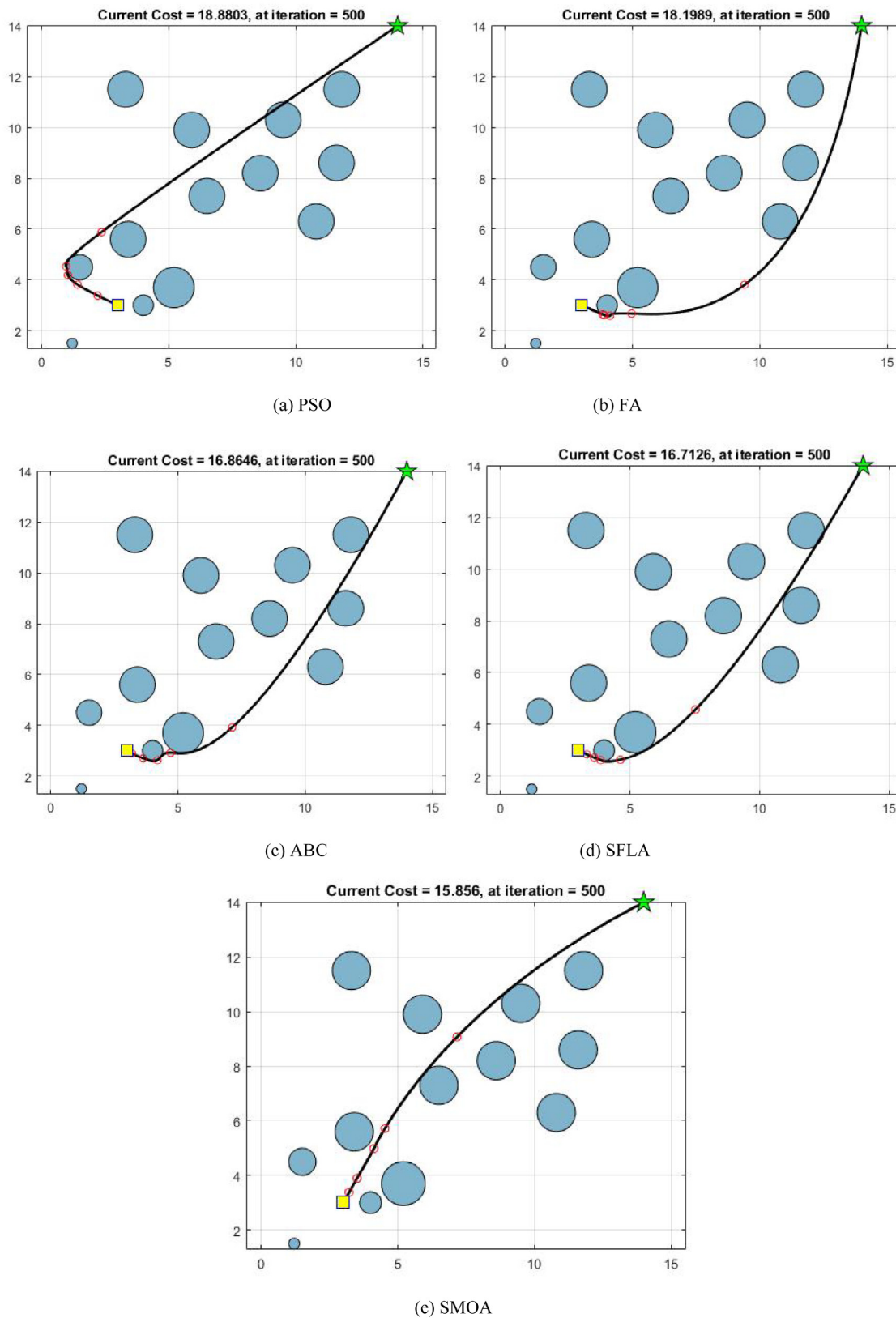


Fig. 4. Map 3 (a) PSO, (b) FA, (c) ABC, (d) SFLA and (e) SMOA.

less time compared to PSO and SFLA approaches, respectively. The comparison between processing time and best optimal paths generated by each approach simulated for this setup is shown in Fig. 10.

The simulation results of each approach's best optimal paths for the fifth environmental setup are presented in Fig. 6(a-e). The values of time taken to generate optimal solutions by each approach are cited in Table 4. Fig. 6(a) shows that PSO gets stuck

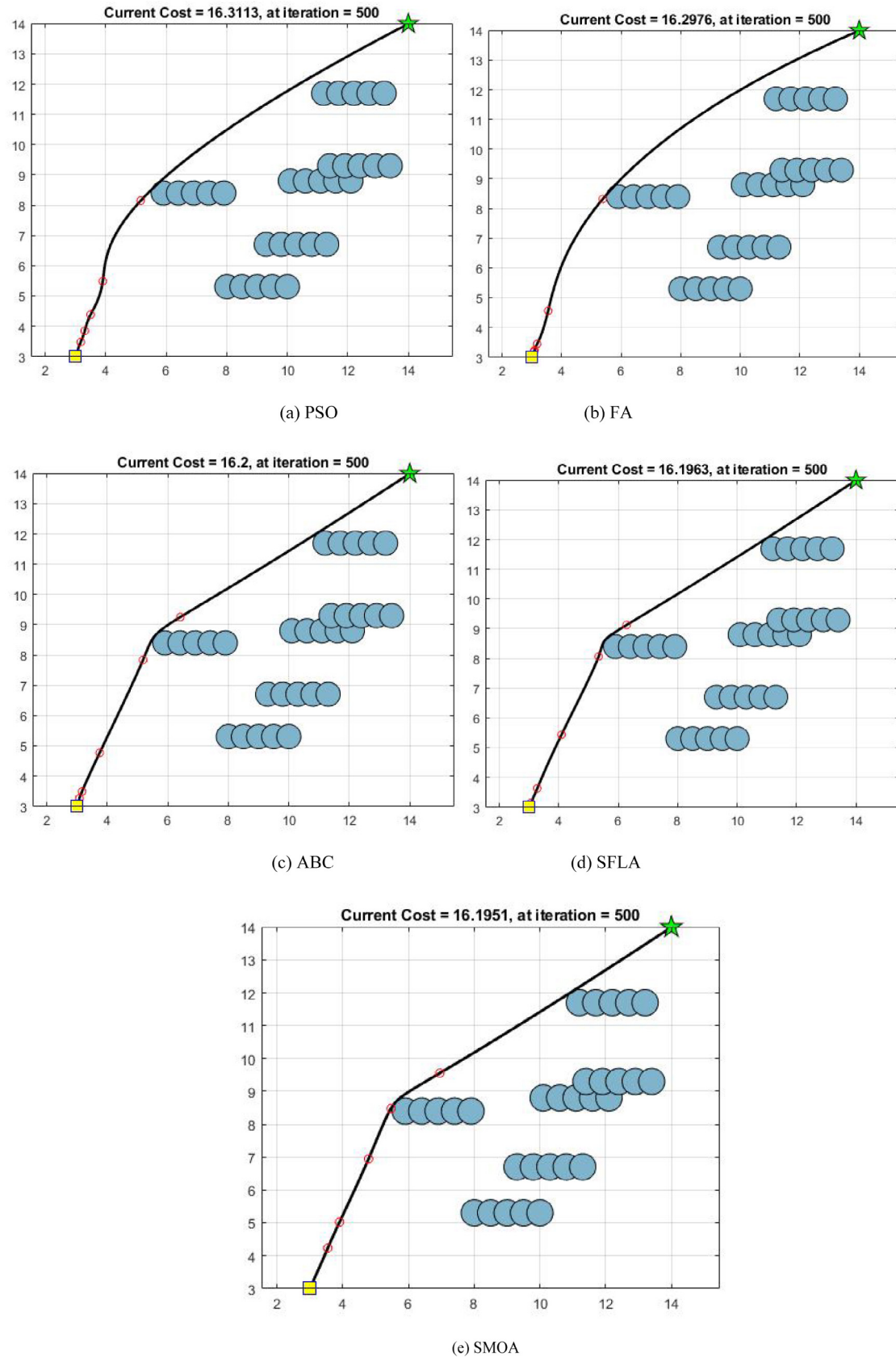


Fig. 5. Map 4 (a) PSO, (b) FA, (c) ABC, (d) SFLA and (e) SMOA.

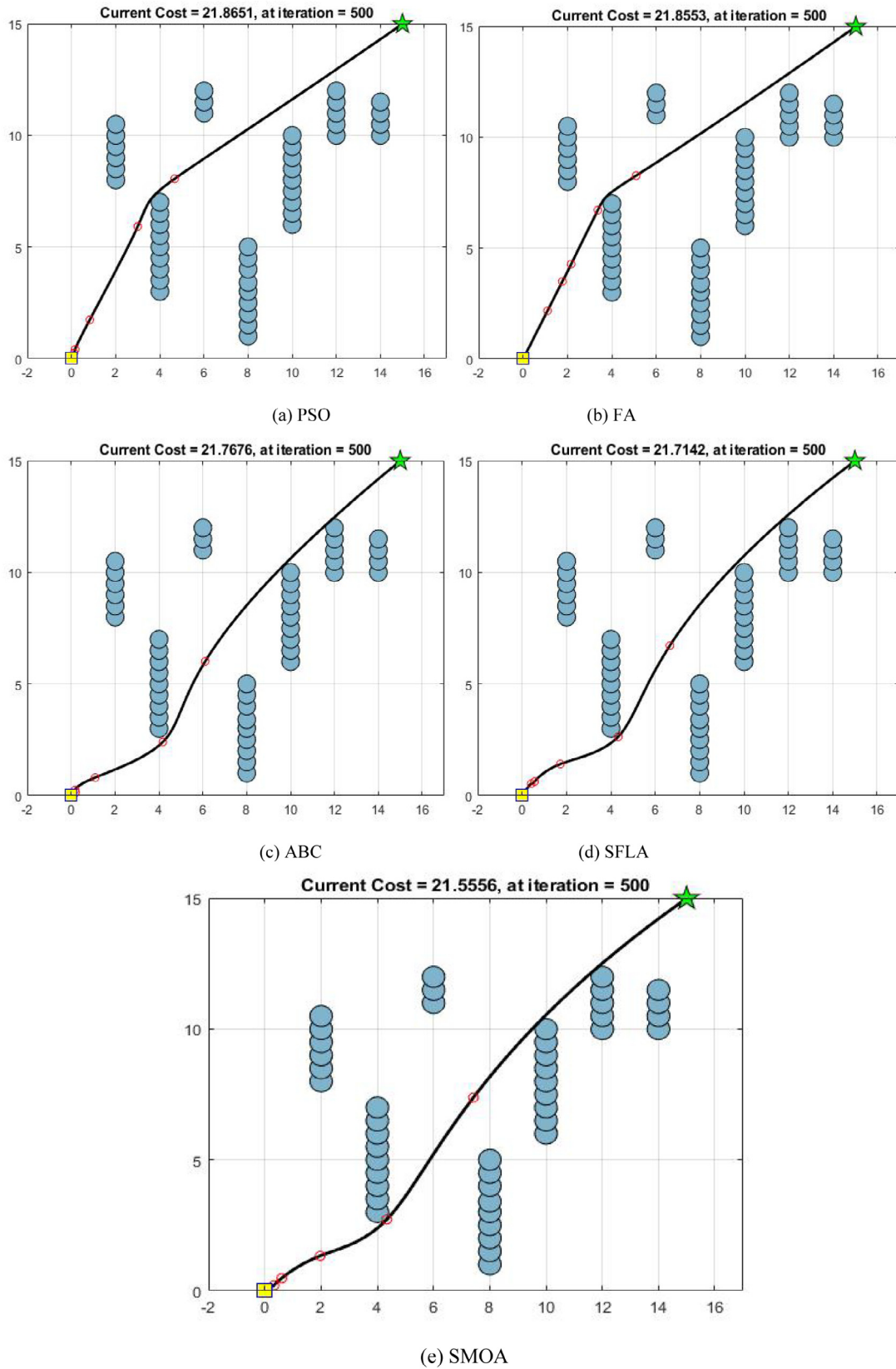


Fig. 6. Map 5 (a) PSO, (b) FA, (c) ABC, (d) SFLA and (e) SMOA.

in a locally optimal solution. It takes the longest route to traverse from the initial to the final point, *i.e.*, 21.8651 units in 226.33 s,

while SMOA takes the shortest collision-free route, *i.e.*, 21.5556 units in 154.26 s. The SMOA approach reduces path length by

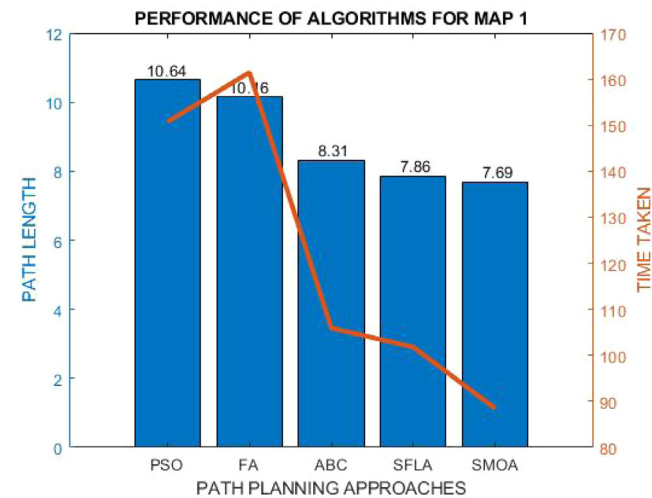


Fig. 7. Performance of approaches for map 1.

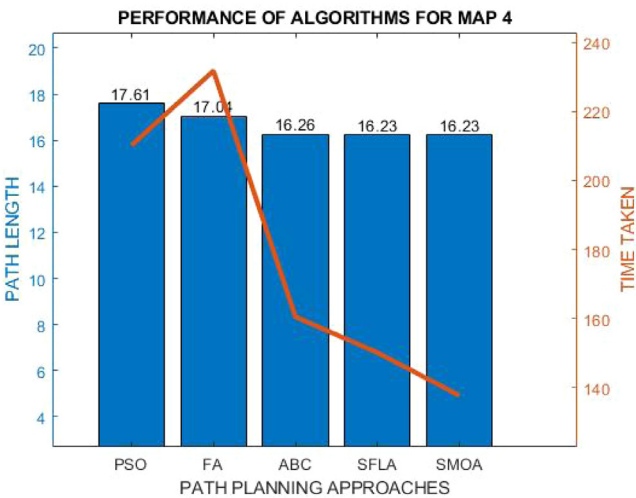


Fig. 10. Performance of approaches for map 4.

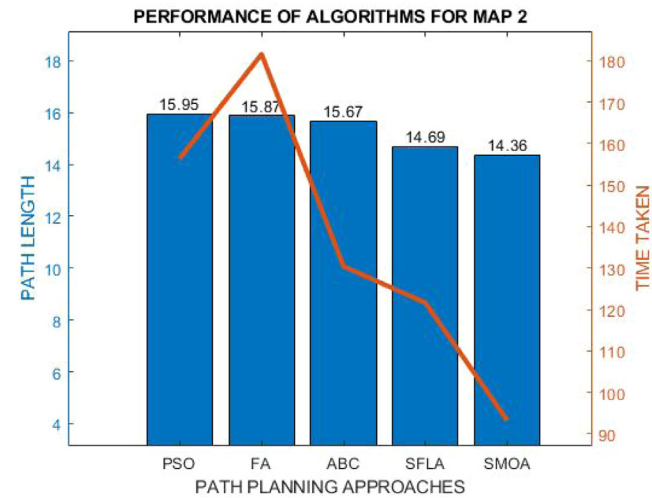


Fig. 8. Performance of approaches for map 2.

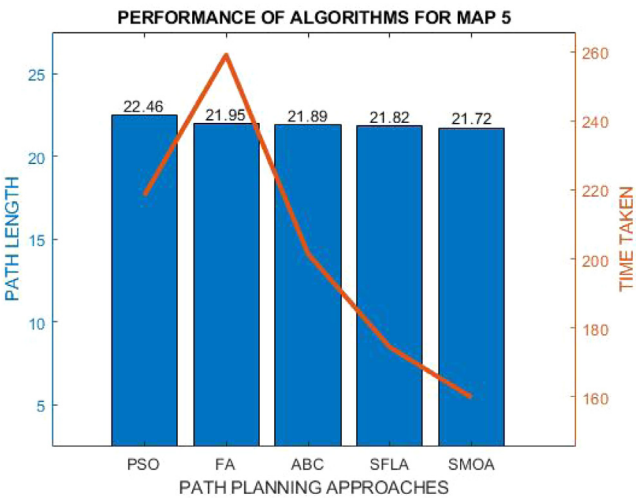


Fig. 11. Performance of approaches for map 5.

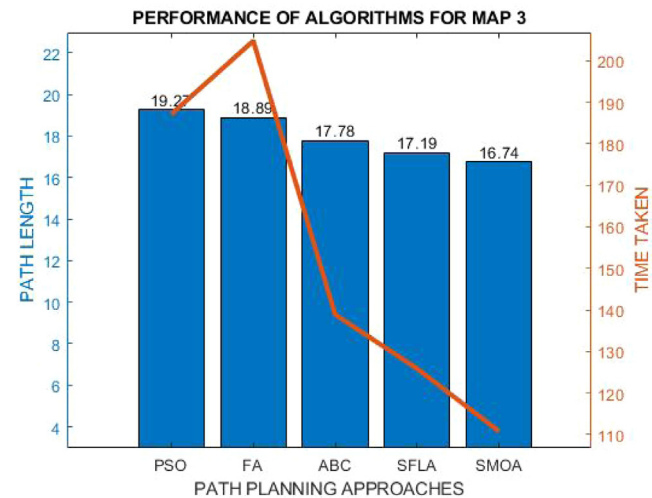


Fig. 9. Performance of approaches for map 3.

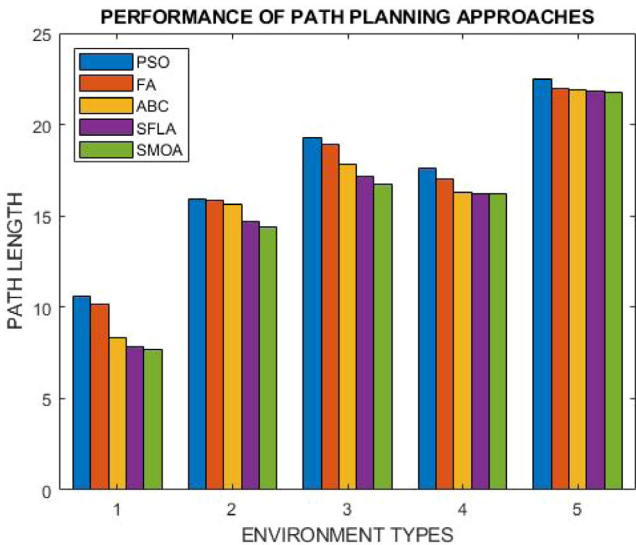


Fig. 12. Comparison Chart of PPOA approaches with respect to path length.

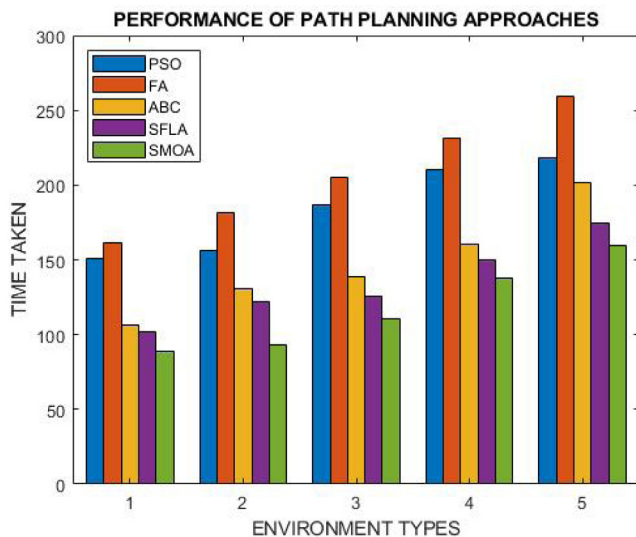


Fig. 13. Comparison Chart of PPOA approaches with respect to processing time.

1.42% and 0.73% compared to PSO and SFLA approaches. The SMOA approach requires 31.84% and 6.12% less time than the PSO and SFLA approach. The comparison between processing time and best optimal paths generated by each approach simulated for this setup is shown in Fig. 11.

Table 4 illustrates that the SMOA approach takes less time to generate a globally optimal collision-free path for all the environmental maps. The values of path length and processing time are precise and repeatable in the SMOA approach, which escalates its adeptness in the application domain. Figs. 12 and 13 illustrates a bar graph showing a comparison of the best values of path length and processing time, respectively, amongst all approaches for all environments. Fig. 12 depicts that SMOA always generates the shortest path, while PSO generates a long length path. The bar graph in Fig. 13 demonstrates that SMOA takes the least processing time while FA takes maximum processing time to generate an optimal path compared to other approaches. Since processing time is the least in the SMOA approach, the planning path's incurring cost would be the least in SMOA. This vindicates competence to implement the SMOA approach for PPOA problems.

7. Conclusion and future scope

This work demonstrates a maiden attempt of a successful implementation of a modified SMOA algorithm for finding the collision-free shortest path in the least time for AMRs. The SMOA approach plans a path by virtually sketching the workspace's spatial configuration and obstacles in its algorithm. The cytoplasm in slime moulds encircles a complete workspace. It follows a subtractive approach (eliminating redundant points and shrinking the cytoplasmic fluid spread) to find a globally optimal collision-free path. For the same, implementation has been done using MATLAB 2020a by taking five environmental maps of different complexity levels. The results demonstrated that modified SMOA not only planned the optimal path without collision, it successfully evaded the local minima as well. In order to see the efficiency of the modified SMOA, the results obtained by this algorithm were compared with those obtained from PSO, FA, ABC and SFLA. The results demonstrated that modified SMOA outperformed other mentioned approaches significantly in producing optimal path with the least cost, in the least processing time and with less computational efforts.

Additionally, the simulation results demonstrated that modified SMOA approach is very consistent and precise approach as the best costs' values are closest to each other for each map, as shown in Table 4. The simulation analysis also determines that this approach is repeatable, efficient and proficient, which must be of interest for application in the PPOA problem. The weight factor in SMOA facilitates a specific disturbance rate to ensure a faster convergence rate and circumvent local minima for all simulation environments. SMOA has better decision-making capabilities and generates the shortest optimal path circumventing obstacles based on information available to it in the least computation time, cost, and effort compared to PSO, FA, ABC and SFLA. So, it can be concluded that SMOA is a reliable approach for the PPOA problem, which has a faster convergence speed and local optima evasion capability. In the future, testing on the real AMR would be carried out in order to validate the authenticity of the simulated results.

CRediT authorship contribution statement

Divya Agarwal: Conceptualization, Data curation, Methodology, Investigation, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Pushendra S. Bharti:** Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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